$$\nu(\tau) = \frac{1}{2}\sum_i c_i^2 \left\langle \{\hat{q}_i(\tau), \hat{q}_i\} \right\rangle_{\hat{\rho}_\mathcal{E}} \equiv \int_0^\infty \mathrm{d}\omega J(\omega) \coth\left(\frac{\omega}{2k_\mathrm{B}T}\right) \cos(\omega\tau)\,, \tag{5.139}$$

$$\eta(\tau) = \frac{\mathrm{i}}{2}\sum_i c_i^2 \left\langle [\hat{q}_i(\tau), \hat{q}_i] \right\rangle_{\hat{\rho}_\mathcal{E}} \equiv \int_0^\infty \mathrm{d}\omega J(\omega) \sin(\omega\tau)\,. \tag{5.140}$$